

Exp. No: 3

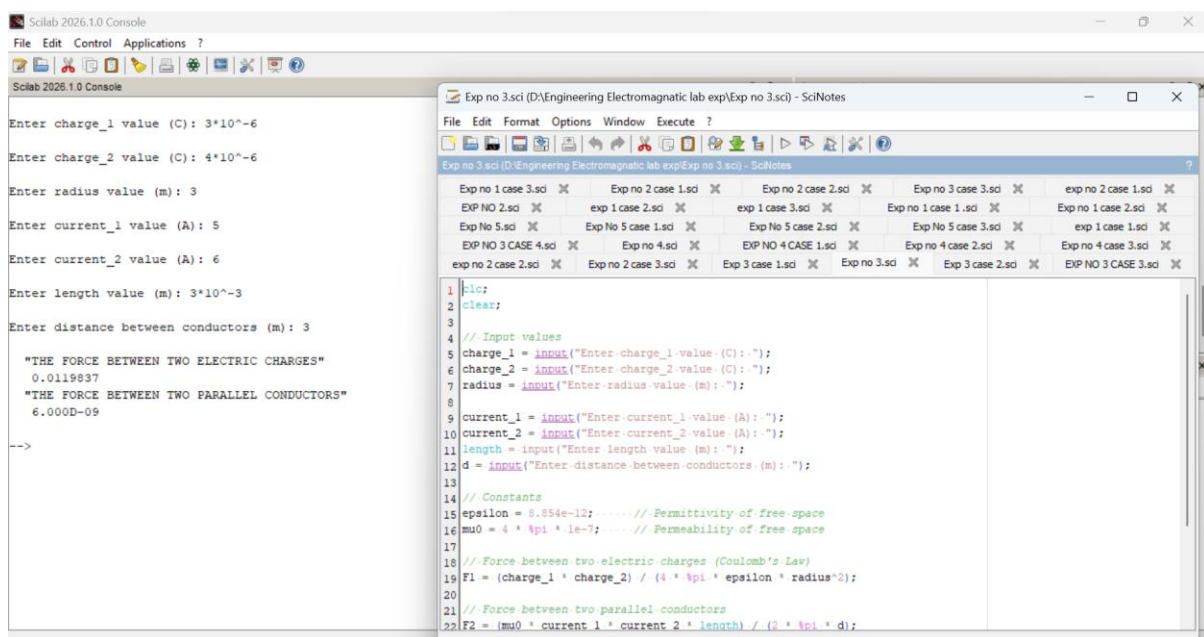
Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance.

Aim:

To Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using SCILAB.

Software required:

- SCILAB version 6.0.1



The screenshot displays the SCILAB 2026.1.0 Console window on the left and the SciNotes window on the right. The Console window shows the execution of a script with the following input values:

```
Enter charge_1 value (C): 3*10^-6
Enter charge_2 value (C): 4*10^-6
Enter radius value (m): 3
Enter current_1 value (A): 5
Enter current_2 value (A): 6
Enter length value (m): 3*10^-3
Enter distance between conductors (m): 3
```

The output of the script is displayed below the inputs:

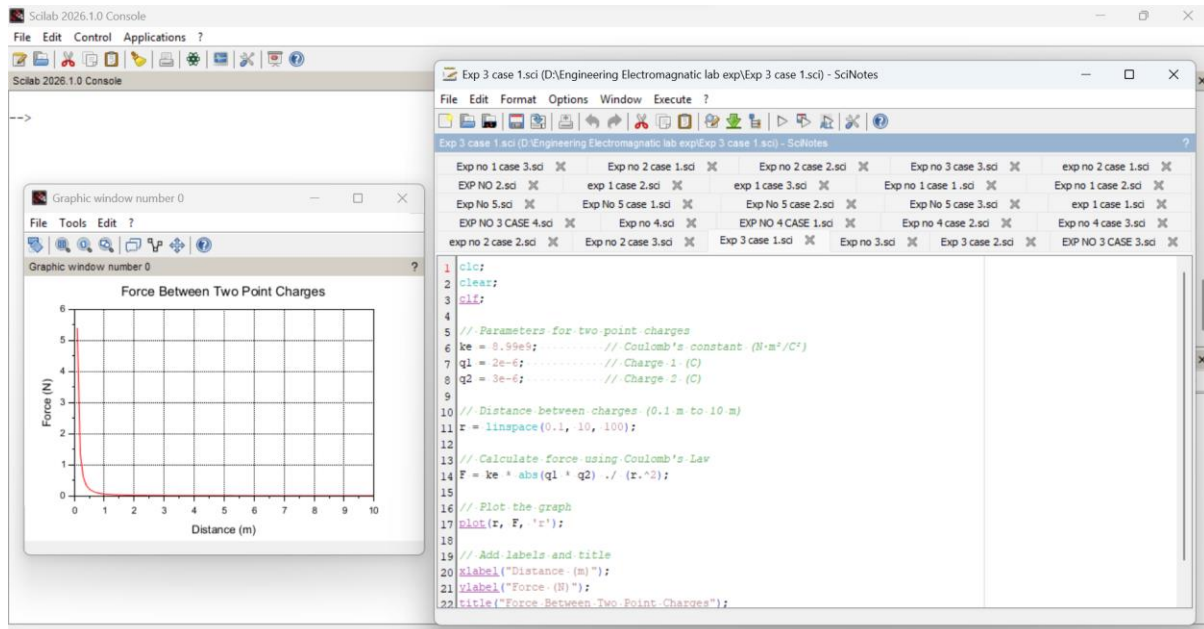
```
"THE FORCE BETWEEN TWO ELECTRIC CHARGES"
0.0119837
"THE FORCE BETWEEN TWO PARALLEL CONDUCTORS"
6.000D-09
-->
```

The SciNotes window shows the source code of the script, which is as follows:

```
1 |clc;
2 |clear;
3 |
4 |// Input values
5 |charge_1 = input("Enter charge_1 value (C): ");
6 |charge_2 = input("Enter charge_2 value (C): ");
7 |radius = input("Enter radius value (m): ");
8 |
9 |current_1 = input("Enter current_1 value (A): ");
10 |current_2 = input("Enter current_2 value (A): ");
11 |length = input("Enter length value (m): ");
12 |d = input("Enter distance between conductors (m): ");
13 |
14 |// Constants
15 |epsilon = 8.854e-12; // Permittivity of free space
16 |mu0 = 4 * pi * 1e-7; // Permeability of free space
17 |
18 |// Force between two electric charges (Coulomb's Law)
19 |F1 = (charge_1 * charge_2) / (4 * pi * epsilon * radius^2);
20 |
21 |// Force between two parallel conductors
22 |F2 = (mu0 * current_1 * current_2 * length) / (2 * pi * d);
```

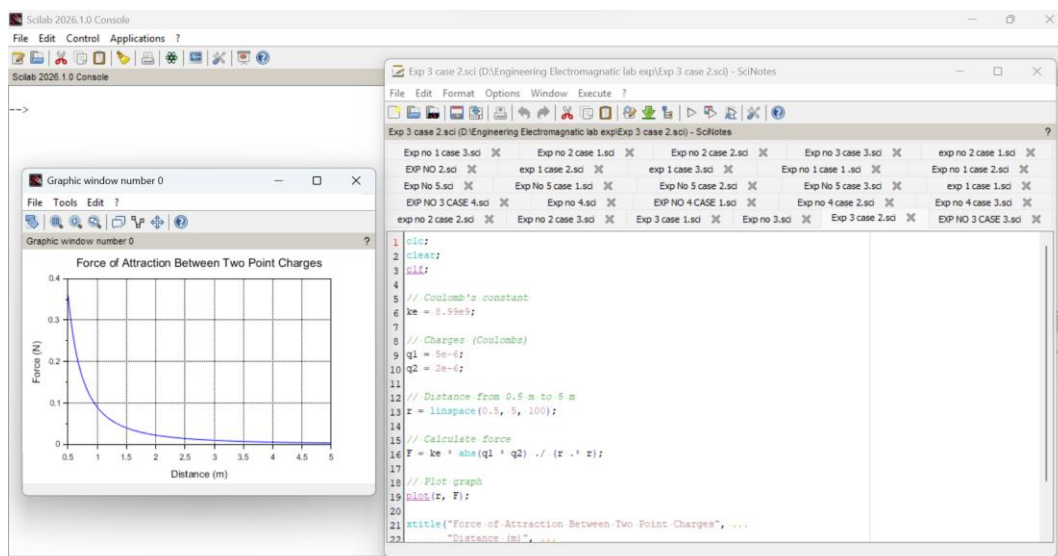
Case 1:

Given two-point charges $q_1=2\times 10^{-6}$ and $q_2=3\times 10^{-6}$ C, calculate and plot the force of attraction between them as a function of distance, ranging from 0.1 m to 10 m. Explain how the force changes with the distance between the charges.



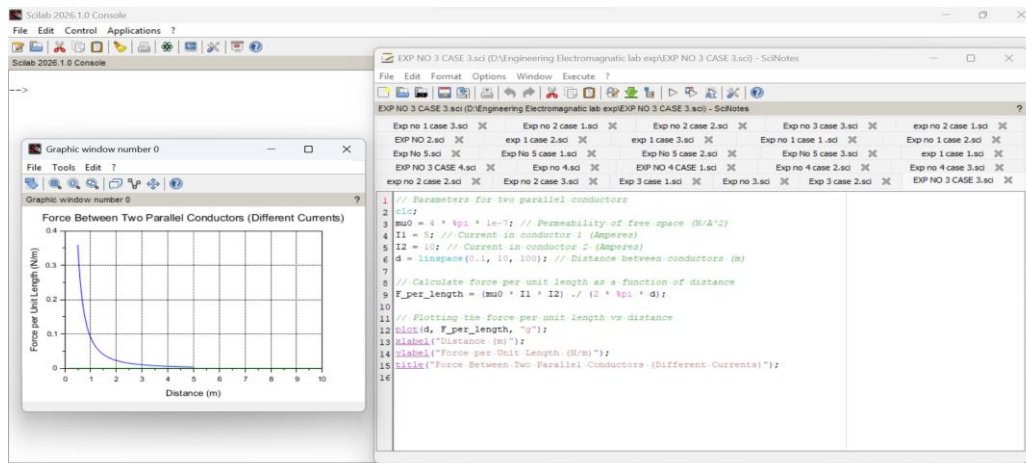
Case 2:

Given two-point charges $q_1=5\times 10^{-6}$ C and $q_2=2\times 10^{-6}$, calculate and plot the force of attraction between them as the distance varies from 0.5 m to 5 m. Discuss how the force behaves with the varying distance and charge magnitudes.



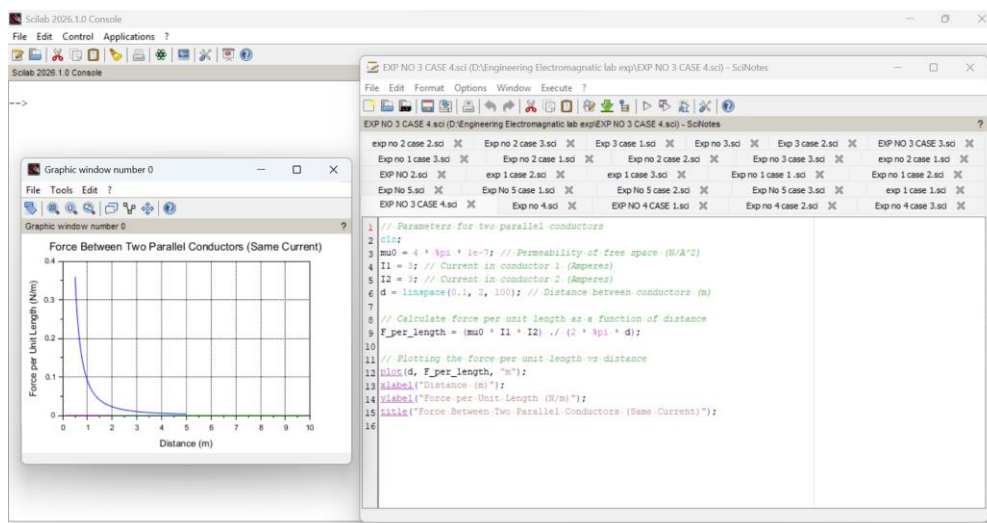
Case 3:

Two parallel conductors are separated by a distance d . The first conductor carries a current of $I_1=5$ A, and the second conductor carries a current of $I_2=10$ A. Calculate and plot the force per unit length between the conductors for varying distances from 0.1 m to 10 m. Explain how the force changes with the distance between the conductors



Case 4:

Given two parallel conductors carrying currents of 3 A, calculate and plot the force per unit length between the conductors for varying distances between 0.1 m to 2 m. Discuss the relationship between the distance and the force between the conductors.



Result : Thus the program was verified for the force of attraction or repulsion acting along a straight line between two electric charges is directly proportional to the product of the charges and inversely to the square of the distance between them using SCILAB was successfully.